Bassez 2021 Breast Cancer — analysis summary

RUN 2 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

41/42 samples intermix on one continuous manifold (per-neighbourhood iLISI median ~9 of 41 after Harmony θ=10 vs 5.6 unintegrated); no per-patient archipelago, 80 island cells removed. Judged on the embedding.

Gate 2 — module geography → GO (non-branching)

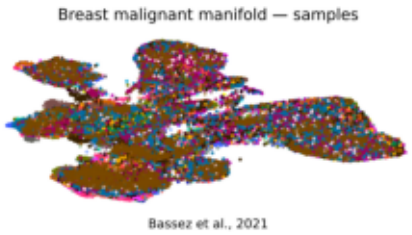
Single pSom→Crest developmental gradient (smoothed pSom↔Crest r = −0.82); Crest terminal, gastrulation weak/flat. A PPT tree fit as an honest branching test put every tip at negative Gast — no Gast arm, no hub.

Gate 3 — hub cross-repression → NA

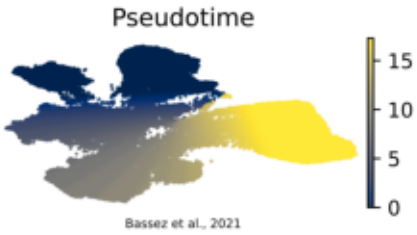
Non-branching manifold — no bifurcation and no Gast/pSom hub, so the cross-repression test does not apply. Crest enriches the trajectory terminus (not a Gast arm): no Crest-arm inversion flag.

STEP FIGURES

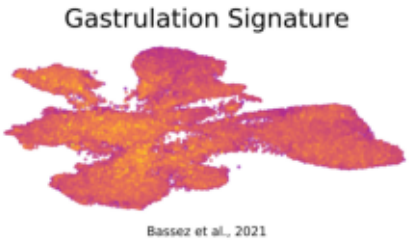
Manifold — samples



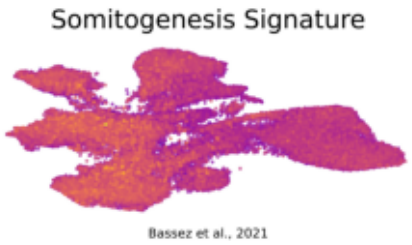
Pseudotime



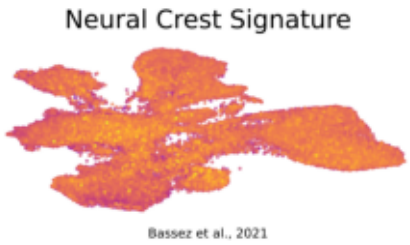
Gastrulation



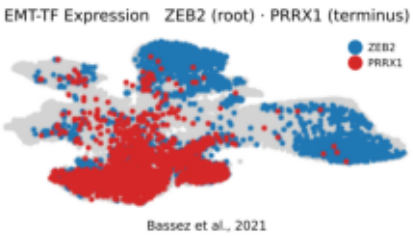
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (Bassez et al., 2021), 10x — 65,802 malignant cells, 41 patients (TNBC / ER+ / HER2+, pre/on anti-PD1).

Verdict.
The embryonic programs reproduce as a NON-BRANCHING single axis: a posterior-somitogenesis origin resolving into a neural-crest terminus (pSom↔Crest r = −0.82), with gastrulation weak and no Gast/pSom hub. Geometry = Linear-pSom/Crest. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TF axis ZEB2 (early) → PRRX1 (late).

Parameter choices

Malignant cells	65,885 → 65,802 (thresholded, de-islanded)	**Sample threshold**	16 cells (multiplicative-gap; drops one 3-cell sample) → 41/42
Overdispersed genes	2,498 (scFates find_overdispersed)	**Integration**	Harmony θ=10, n_pcs=20 (iLISI-sweep derived)
Embedding	UMAP spread 1.3, n_neighbors 30, seed 1; island removal 80 cells / 21 specks	**Trajectory**	ElPiGraph principal CURVE, 20 nodes (fold-free, self-approach 2.43)
Root	pSom-high / Crest-low tip (origin; Crest late)	**EMT-TFs**	ZEB2 (root) · PRRX1 (terminus), detection-gated